Supplementary Material

Photovoltaic Power Nowcasting with LightGBM

Saumya Sinha, Priyansh Yadav

Companion to the main manuscript. Section, table and equation numbers here carry the prefix S; unprefixed numbers refer to the main manuscript.

S1 Inputs and Sample Counts

Inputs. The benchmark files `opera_dataset_1w_10min.csv` (24,031 rows) and `opera_dataset_3w_90min.csv` (24,024 rows). For windows $w \in \{10, 20, 60\}$ min, statistics $s \in \{\text{mean}, \max, \min, \text{std}\}$ and signals $v \in \{\texttt{Rad}, \texttt{Tamb}, \texttt{Tmod}\}$, the environmental feature count is $d_{\text{env}} = 3 \times 4 \times 3 = 36$. The target is $y = \texttt{Pa1}$; current, voltage and power-named columns are excluded.

Alignment. A 90-minute sample at time t is kept only if the nine trailing 10-minute steps $t, t-10, \ldots, t-80$ min exist and are consecutive, giving $n = 23{,}984$.

Splits (scikit-learn convention: test size $= \lceil pn \rceil$):

$$n_{\text{test}} = \lceil 0.20 \cdot 23984 \rceil = 4797, \qquad n_{\text{outer}} = 23984 - 4797 = 19187, \tag{S1}$$

$$n_{\text{val}} = \lceil 0.20 \cdot 19187 \rceil = 3838, \qquad n_{\text{fit}} = 19187 - 3838 = 15349. \tag{S2}$$

Seed 42 gives the outer split, seed 43 the validation split.

S2 Feature Construction

Let $m \in \{0, \ldots, 1439\}$ be the minute of day, $d \in \{1, \ldots, 365\}$ the day of year, and $G_k, T_{\text{mod},k}, T_{\text{amb},k}$ the mean irradiance, module and ambient temperature of window k ($k = 1$ most recent). The ten engineered features are

$$x_{1,2} = \sin, \cos \tfrac{2\pi m}{1440}, \qquad x_{3,4} = \sin, \cos \tfrac{2\pi d}{365.25}, \qquad x_5 = G_1^2, \qquad x_6 = G_1 T_{\text{mod},1}, \tag{S3}$$

$$x_7 = G_1 T_{\text{amb},1}, \qquad x_8 = T_{\text{mod},1} - T_{\text{amb},1}, \qquad x_9 = G_1 - G_2, \qquad x_{10} = G_1 - G_3, \tag{S4}$$

so the full model has $d = 36 + 10 = 46$ inputs. The cyclic encoding places a phase on the unit circle, with chord distance $2|\sin(\pi\Delta/T)|$; e.g. 23:50 and 00:00 are 0.0436 apart instead of 1,430 minutes.

S3 Gradient Boosting Derivation

Additive model. $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x})$, $F_0 = \arg\min_c \sum_i L(y_i, c)$ (mean for squared error, median for absolute error).

Second-order objective. With $g_i = \partial L/\partial F$ and $\eta_i = \partial^2 L/\partial F^2$ at $F_{m-1}(\mathbf{x}_i)$, a tree with J leaves, leaf outputs w_j and penalty $\frac{1}{2}\lambda_2 w_j^2 + \lambda_1 |w_j|$ gives

$$\mathcal{L}_m \simeq \sum_{j=1}^{J} \Big[G_j w_j + \tfrac{1}{2}(H_j + \lambda_2) w_j^2 + \lambda_1 |w_j| \Big], \quad G_j = \sum_{i \in I_j} g_i, \; H_j = \sum_{i \in I_j} \eta_i. \tag{S5}$$

Minimising each term yields

$$w_j^* = -\frac{\mathcal{T}_{\lambda_1}(G_j)}{H_j + \lambda_2}, \quad \mathcal{T}_{\lambda_1}(G) = \text{sign}(G) \max(|G| - \lambda_1, 0), \quad \mathcal{L}_m^* = -\tfrac{1}{2} \sum_j \frac{\mathcal{T}_{\lambda_1}(G_j)^2}{H_j + \lambda_2}. \tag{S6}$$

A split of a node (G, H) into (G_L, H_L), (G_R, H_R) has gain

$$\text{Gain} = \tfrac{1}{2} \Big[\tfrac{\mathcal{T}_{\lambda_1}(G_L)^2}{H_L + \lambda_2} + \tfrac{\mathcal{T}_{\lambda_1}(G_R)^2}{H_R + \lambda_2} - \tfrac{\mathcal{T}_{\lambda_1}(G)^2}{H + \lambda_2} \Big], \tag{S7}$$

and leaf-wise growth splits the leaf of largest gain until the configured number of leaves is reached. For $L = \frac{1}{2}(y - F)^2$: $g_i = F_{m-1}(\mathbf{x}_i) - y_i$, $\eta_i = 1$. For $L = |y - F|$: $g_i = \text{sign}(F_{m-1}(\mathbf{x}_i) - y_i)$ with constant substituted curvature (library objective `regression_l1`; squared error is `regression`).

Early stopping and refit. Each candidate is fitted on the fitting part and the validation MAE ℓ_m is evaluated each round. Training stops when $\ell_m > \min_{k<m} \ell_k$ for 80 consecutive rounds; $m^* = \arg\min_m \ell_m$. For the selected model $m^* = 505 < 1400$, so early stopping ended training. The configuration is then refitted on the outer training set with exactly m^* trees and evaluated once on the test set.

S4 Evaluation Metrics

For N test samples with residuals $e_i = y_i - \hat{y}_i$ and mean $\bar{y}$:

$$\mathrm{MAE} = \tfrac{1}{N}\sum_i |e_i|, \qquad \mathrm{MSE} = \tfrac{1}{N}\sum_i e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \tag{S8}$$

$$\mathrm{SSE} = \sum_i e_i^2, \quad \mathrm{SST} = \sum_i (y_i - \bar{y})^2 = N s_y^2, \qquad R^2 = 1 - \mathrm{SSE}/\mathrm{SST} = 1 - \mathrm{MSE}/s_y^2. \tag{S9}$$

By Jensen's inequality MAE $\leq$ RMSE; for Gaussian errors MAE $= \sqrt{2/\pi}\,\mathrm{RMSE} \approx 0.798\,\mathrm{RMSE}$, whereas here $88.67/228.02 = 0.389$, indicating heavy-tailed errors (maximum $|e_i| = 3{,}982$ W), which is why both metrics are reported.

Worked values ($N = 4797$ rows of `random_test_predictions.csv`, columns `actual_power_w` and `selected_model_prediction_w`; the prediction equals that of the single model `engineered__LGBM_RMSE_127`):

$$\textstyle\sum_i |e_i| = 425{,}367.6 \text{ W}, \qquad \mathrm{MAE} = 425367.6/4797 = 88.674 \text{ W}, \tag{S10}$$

$$\mathrm{SSE} = 249{,}406{,}480 \text{ W}^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{SSE}/N} = 228.018 \text{ W}, \tag{S11}$$

$$s_y^2 = 77{,}125{,}240 \text{ W}^2 \ (s_y = 8782.1 \text{ W}), \qquad R^2 = 1 - \mathrm{SSE}/\mathrm{SST} = 0.999326. \tag{S12}$$

Consistency check from the rounded values: $s_y^2 = \mathrm{MSE}/(1-R^2) \approx 77{,}141{,}128$ ($s_y \approx 8783$ W), a 0.01% rounding difference.

Percentage change. Relative to a reference value m_{ref},

$$\Delta(m) = 100\,\frac{m_{\mathrm{ref}} - m}{m_{\mathrm{ref}}} \qquad (\Delta > 0 \Rightarrow \text{lower error}). \tag{S13}$$

Against Random Forest, 90 min (173.47 W, 360.13 W): $\Delta(\mathrm{MAE}) = 100 \cdot 84.80/173.47 = 48.9\%$, $\Delta(\mathrm{RMSE}) = 100 \cdot 132.11/360.13 = 36.7\%$. Against Araujo (354.81, 641.36): 75.0% and 64.4%. Against 3CNN+2LSTM, 10 min (274.87, 531.08): 67.7% and 57.1%. Benchmark MAE/RMSE are copied from the published tables of the reference papers.

S5 Model Selection Score

Candidates are ranked on the validation subset by

$$S = 0.55\,\frac{\mathrm{MAE}_{\mathrm{val}}}{173.47} + 0.45\,\frac{\mathrm{RMSE}_{\mathrm{val}}}{360.13}, \tag{S14}$$

where 173.47 W and 360.13 W are benchmark scale constants making both errors dimensionless; the lowest S is selected and test labels are not used. Worked example (engineered, 127 leaves; $\mathrm{MAE}_{\mathrm{val}} = 99.47$, $\mathrm{RMSE}_{\mathrm{val}} = 266.92$): $0.55 \times 0.5734 + 0.45 \times 0.7412 = 0.3154 + 0.3335 = 0.6489$. The table reports all eight validation R^2,

Table S1: Validation performance and selection scores of the eight LightGBM candidates. The selected candidate is shaded; the best iteration is selected by validation, and a value equal to the configured maximum indicates that the maximum was reached.

Feature set	Configuration	R^2_{val}	MAE (W)	RMSE (W)	Best iteration	MAE/173.47	RMSE/360.13	Score S
Environmental	Sq. error, 31 leaves	0.998727	146.35	311.22	1800	0.8437	0.8642	0.8529
Environmental	Sq. error, 63 leaves	0.998661	151.46	319.20	470	0.8731	0.8863	0.8791
Environmental	Sq. error, 127 leaves	0.998641	148.69	321.54	522	0.8572	0.8928	0.8732
Environmental	Abs. error, 31 leaves	0.998383	151.94	350.76	1800	0.8759	0.9740	0.9200
Engineered	Sq. error, 31 leaves	0.999079	107.40	264.75	652	0.6191	0.7352	0.6713
Engineered	Sq. error, 63 leaves	0.999071	102.39	265.91	472	0.5902	0.7384	0.6569
Engineered	Sq. error, 127 leaves	0.999064	99.47	266.92	505	0.5734	0.7412	0.6489
Engineered	Abs. error, 31 leaves	0.998940	107.24	284.09	1800	0.6182	0.7889	0.6950

MAE, RMSE, best-iteration and selection-score values. Early stopping and candidate ranking use the same validation subset, so these scores are used for model selection rather than as independent estimates of test performance.

S6 Reproduction Procedure

1. Load both CSV files, align, keep $n = 23{,}984$ samples (S1).

2. Split with seed 42 (outer 19,187 / test 4,797), then seed 43 (fit 15,349 / validation 3,838).
3. For each feature set and configuration of Table S3, fit with early stopping; record MAE_{val}, RMSE_{val}, R^2_{val}, m^*.
4. Compute S and select the minimum (S5).
5. Refit with m^* trees on the outer training set; predict the test set once and save the predictions.
6. Run `verify_results.py` (metrics, counts) and `compute_derived.py` (percentage changes) on `results/`.

The saved model also records `random_state`=42, `n_jobs`=2, `min_child_weight`=0.001, `min_split_gain`=0, `boosting_type`=gbdt.

Table S2: Agreement between the main manuscript and the recomputed values.

Quantity	Reported	Recomputed	Basis
Test / outer-training size	4,797 / 19,187	4,797 / 19,187	S1
Features (env. / engineered / total)	36 / 10 / 46	36 / 10 / 46	S1, S2
MAE, RMSE, R^2 (holdout)	88.67, 228.02, 0.999326	88.6737, 228.0179, 0.9993259	S4
ΔMAE, ΔRMSE vs. RF 90 min	48.9, 36.7%	48.88, 36.68%	S4
Δ vs. Araujo	75.0, 64.4%	75.01, 64.45%	S4
Δ vs. 3CNN+2LSTM (10 min)	67.7, 57.1%	67.74, 57.06%	S4
Selection scores (8 candidates)	0.6489 (selected), ...	identical at 4 d.p.	S5 / Table S1
Errors below 100 W / 500 W	75.3 / 97.3%	75.32 / 97.27%	S4
Largest absolute error	3,982 W	3,981.8 W	S4

S7 Candidate Configurations and Notation

Table S3: Candidate LightGBM configurations evaluated for each feature set. All use `max_depth`= −1, `subsample_freq`= 1 and seed 42; trees are the configured maximum, reduced by early stopping (patience 80).

Configuration	Objective	Max. trees	ν	Leaves	Min. samples/leaf	Row subs.	Col. subs.	λ_2	λ_1
31 leaves	squared error	1800	0.025	31	25	0.90	0.9	3.0	0.05
63 leaves	squared error	1600	0.025	63	40	0.90	0.9	5.0	0.10
127 leaves (selected)	squared error	1400	0.030	127	60	0.85	0.9	8.0	0.10
31 leaves, abs. error	absolute error	1800	0.025	31	25	0.90	0.9	3.0	0.05

Table S4: Notation.

Symbol	Meaning	Symbol	Meaning
`Pa1`	Measured output power (target), W	h_m, m^*	Tree at iteration m; best iteration
G, T_{amb}, T_{mod}	Irradiance, ambient and module temperature	g_i, η_i	First/second loss derivative at sample i
$y_i, \hat{y}_i, e_i$	Measurement, prediction, residual	G_j, H_j	Sums of g_i, η_i over leaf j
N, n	Test samples (4,797); aligned samples (23,984)	λ_1, λ_2	L_1, L_2 leaf penalties
S	Validation selection score	w_j^*	Optimal leaf output
ν	Learning rate	$\Delta(\cdot)$	Percentage change vs. reference
s_y^2	Test-set target variance		

S8 Methodological Checklist

- *Applied*: complete-window alignment (all nine trailing 10-minute steps present).
- *Applied*: predictors exclude output power, current and voltage.
- *Applied*: model selection and early stopping on a validation subset of the training data; test rows not used for selection.
- *Applied*: the selected model is evaluated once on the held-out test set.
- *Not applied*: chronological or blocked evaluation, nested hyperparameter selection, multiple seeds, uncertainty intervals, test-set feature-group ablation, validation on an independent site or season.